...0632 이진아.

$\mu = GM$

1) Newton's Law, $\ddot{\vec r} = -\frac{\mu}{r^3}\vec r + \frac{\vec F}{m}$

$\vec r = x i + y j + z k$, $\vec F = F_x i + F_y j + F_z k$

$\vec x = [x, \dot x, y, \dot y, z, \dot z]^T$. (state)

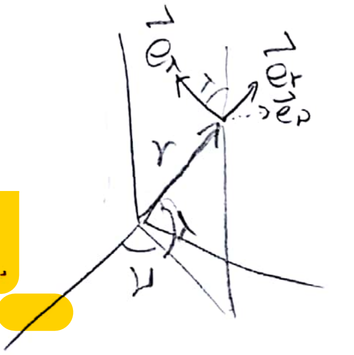
By a, $\ddot x i + \ddot y j + \ddot z k = -\frac{\mu}{\sqrt{x^2+y^2+z^2}^3}(x i + y j + z k) + \frac{1}{m}(F_x i + F_y j + F_z k)$

$$\Rightarrow \begin{cases} \ddot x = -\frac{\mu}{(\sqrt{x^2+y^2+z^2})^3}x + \frac{F_x}{m} \\ \ddot y = -\frac{\mu}{(\sqrt{x^2+y^2+z^2})^3}y + \frac{F_y}{m} \\ \ddot z = -\frac{\mu}{(\sqrt{x^2+y^2+z^2})^3} + \frac{F_z}{m} \end{cases}$$

$$\Rightarrow \begin{bmatrix} \dot x \\ \ddot x \\ \dot y \\ \ddot y \\ \dot z \\ \ddot z \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -w^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -w^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -w^2 & 0 \end{bmatrix}}_{\equiv A \text{ (system matrix)}} \begin{bmatrix} x \\ \dot x \\ y \\ \dot y \\ z \\ \dot z \end{bmatrix} + \overbrace{\begin{bmatrix} 0 & 0 & 0 \\ \frac{1}{m} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & \frac{1}{m} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{1}{m} \end{bmatrix}}^{\equiv B \text{ (control matrix)}} \underbrace{\begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix}}_{U \text{ (control)}}$$

where $w^2 = \frac{\mu}{(x^2+y^2+z^2)^{3/2}}$

$\Rightarrow \dot X = AX + BU$.




2) $\vec r = r\vec e_r + \nu \vec e_\nu + \lambda \vec e_\lambda$, $\vec F = F_r \vec e_r + F_\nu \vec e_\nu + F_\lambda \vec e_\lambda$

$\Rightarrow \vec\omega = -\dot\lambda \vec e_r + \dot\nu \cos\lambda \vec e_\nu + \dot\nu \sin\lambda \vec e_\lambda$   ~/5

$\dot{\vec r} = \dot r \vec e_r + \dot\nu \vec e_\nu + \dot\lambda \vec e_\lambda + r\dot{\vec e}_r + \nu \dot{\vec e}_\nu + \lambda \dot{\vec e}_\lambda$

$= \dot r \vec e_r + \dot\nu \vec e_\nu + \dot\lambda \vec e_\lambda + \vec\omega \times (r\vec e_r + \nu \vec e_\nu + \lambda \vec e_\lambda)$ By Coriolis thm.

$= \dot r \vec e_r + \dot\nu \vec e_\nu + \dot\lambda \vec e_\lambda + (-\dot\lambda \vec e_r + \dot\nu\cos\lambda \vec e_\nu + \dot\nu \sin\lambda \vec e_\lambda)\times(r\vec e_r + \nu\vec e_\nu + \lambda \vec e_\lambda)$

$= (\dot r + \lambda\dot\nu\cos\lambda - \nu\dot\nu\sin\lambda)\vec e_r + (\dot\nu + r\dot\lambda + \dot\nu r \overset{}{\underset{\sin\lambda}{}})\vec e_\nu + (\dot\lambda - \dot\lambda\nu - r\dot\nu\cos\lambda)\vec e_\lambda$

$\ddot{\vec r} = (\ddot r + \dot\lambda\dot\nu\cos\lambda + \lambda\ddot\nu\cos\lambda - \lambda\dot\nu\sin\lambda - \dot\nu^2\sin\lambda - \nu\ddot\nu\sin\lambda - \nu\dot\nu\cos\lambda)\vec e_r$

$+ (\ddot\nu + \ddot\lambda\lambda + \dot\lambda^2 + \overset{\sin\lambda}{\ddot\nu r} + \underset{\sin\lambda + \dot\nu r\cos\lambda}{\dot\nu\dot r})\vec e_\nu + (\ddot\lambda - \ddot\lambda\nu + \dot\lambda\dot\nu - \dot r\dot\nu\cos\lambda - r\ddot\nu\cos\lambda + r\dot\nu\sin\lambda)\vec e_\lambda$

$+ \vec\omega \times \dot{\vec r}$   By Coriolis thm.

Scanned by CamScanner

2324032 이진아.



$\vec{\omega}\times\dot{\vec r} = (-\dot\lambda\vec e_r + \dot\nu\cos\lambda\vec e_\nu + \dot\nu\sin\lambda\vec e_\lambda)\times$

$\{(\dot r + \lambda\dot\nu\cos\lambda - \nu\dot\nu\sin\lambda)\vec e_r + (\dot\nu + \lambda\dot\lambda + \dot\nu r\underset{\sin\lambda}{\wedge})\vec e_\nu + (\dot\lambda - \lambda\nu - r\dot\nu\cos\lambda)\vec e_\lambda\}$

$= (\dot\nu\dot\lambda\cos\lambda - \dot\nu\nu\lambda\cos\lambda - \nu\dot\nu^2\cos^2\lambda - \dot\nu^2\sin\lambda - \dot\nu\lambda\dot\lambda\sin\lambda - \dot\nu^2 r\sin^2\lambda)\vec e_r$

$(\dot\lambda^2 - \dot\lambda^2\nu - \lambda\dot\lambda\dot\nu\cos\lambda + \dot r\dot\nu\sin\lambda + \dot\nu^2\lambda\sin\lambda\cos\lambda - \nu\dot\nu^2\sin^2\lambda)\vec e_\nu +$

$(-\dot\lambda\dot\nu - \dot\lambda^2\lambda - \dot\lambda\dot\nu r\underset{\sin\lambda}{\wedge} - \dot r\dot\nu\cos\lambda - \lambda\dot\nu^2\cos^2\lambda + \nu\dot\nu^2\sin\lambda\cos\lambda)\vec e_\lambda$

$\ddot{\vec r} = (\ddot r + \dot\nu\dot\lambda\cos\lambda + \ddot\nu\lambda\cos\lambda - \dot\nu\lambda\sin\lambda - \nu\dot\nu^2\sin\lambda - \nu\ddot\nu\sin\lambda - \nu\dot\nu\cos\lambda + \dot\nu\dot\lambda\cos\lambda$

$-\dot\nu\nu\lambda\cos\lambda - \nu\dot\nu^2\cos^2\lambda - \dot\nu^2\sin\lambda - \dot\nu\lambda\dot\lambda\sin\lambda - \dot\nu^2 r\sin^2\lambda)e_r$

$+(\ddot\nu + \ddot\lambda\lambda + \dot\lambda^2 + r\ddot\nu\overset{\sin\lambda}{\wedge} + \dot r\dot\nu\underset{\sin\lambda}{\wedge} + \dot\lambda^2 - \nu\dot\lambda^2 - r\dot\nu\dot\lambda\cos\lambda + \dot r\dot\nu\sin\lambda + \dot\nu^2\lambda\sin\lambda\cos\lambda$

$-\nu\dot\nu^2\sin^2\lambda)\vec e_\nu$ $\underset{+\dot\nu r\cos\lambda}{\wedge}$

$+(\ddot\lambda - \ddot\lambda\nu + \dot\lambda\dot\nu - r\ddot\nu\cos\lambda - \dot r\dot\nu\cos\lambda + r\dot\nu\sin\lambda - \dot\nu\dot\lambda - \dot\lambda^2\lambda - r\dot\nu\dot\lambda\sin\lambda$

$-\dot r\dot\nu\cos\lambda - \lambda\dot\nu^2\cos^2\lambda + \nu\dot\nu^2\sin\lambda\cos\lambda)\vec e_\lambda$

$= \cancel{\ddot r + 2\dot\nu\dot\lambda\cos\lambda + \ddot\nu\lambda\cos\lambda - r\dot\nu^2 - \dot\nu\lambda\sin\lambda - \dot\nu\sin\lambda - \nu\ddot\nu\sin\lambda - \nu\dot\nu\cos\lambda}$

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$(-\dot\lambda\vec e_r + \dot\nu\cos\lambda\vec e_\nu + \dot\nu\sin\lambda\vec e_\lambda)\times\vec e_r$

$= -\dot\nu\cos\lambda\vec e_\lambda + \dot\nu\sin\lambda e_\nu = \dot\nu\sin\lambda\vec e_\nu - \dot\nu\cos\lambda\vec e_\lambda$

$(-\dot\lambda\vec e_r + \dot\nu\cos\lambda\vec e_\nu + \dot\nu\sin\lambda\vec e_\lambda)\times\vec e_\nu$

$= -\dot\nu\sin\lambda\vec e_r - \dot\lambda\vec e_\lambda$

$(-\dot\lambda\vec e_r + \dot\nu\cos\lambda\vec e_\nu + \dot\nu\sin\lambda\vec e_\lambda)\times\vec e_\lambda$

$= \dot\nu\cos\lambda\vec e_r + \dot\lambda\vec e_\nu.$

$\dot{\vec r} = (\dot r - \nu\dot\nu\sin\lambda + \dot\nu\lambda\cos\lambda)\vec e_r +$

$(\dot\nu + \dot\nu r\sin\lambda + \lambda\dot\lambda)\vec e_\nu +$

$(\dot\lambda - \dot\nu r\cos\lambda - \nu\dot\lambda)\vec e_\lambda$

$\ddot r: \{\ddot r - \dot\nu^2\sin\lambda - \nu\ddot\nu\sin\lambda - \nu\dot\nu\cos\lambda + \ddot\nu\lambda\cos\lambda + \dot\nu\dot\lambda\cos\lambda - \dot\nu\lambda\sin\lambda$

$-(\dot\nu + \dot\nu r\sin\lambda + \lambda\dot\lambda)\dot\nu\sin\lambda + (\dot\lambda - \dot\nu r\cos\lambda - \nu\dot\lambda)\dot\nu\cos\lambda\}$

$= \ddot r - \dot\nu^2\sin\lambda - \nu\ddot\nu\sin\lambda - \nu\dot\nu\cos\lambda + \ddot\nu\lambda\cos\lambda + \dot\nu\dot\lambda\cos\lambda - \dot\nu\lambda\sin\lambda$

$-\dot\nu^2\sin\lambda - \dot\nu^2 r\sin^2\lambda - \lambda\dot\lambda\dot\nu\sin\lambda + \dot\lambda\dot\nu\cos\lambda - \dot\nu^2 r\cos^2\lambda - \nu\dot\nu\dot\lambda\cos\lambda$

$= (-\dot\lambda \vec{e}_r + \dot\nu \cos\lambda \vec{e}_\nu + \dot\nu \sin\lambda \vec{e}_\lambda) \times$

$\{(\dot r + \lambda\dot\nu\cos\lambda - \nu\dot\nu\sin\lambda)\vec{e}_r + (\dot\nu + \lambda\dot\lambda + \dot\nu r\sin\lambda)\vec{e}_\nu$
$+ (\dot\lambda - \dot\lambda\nu - \dot\nu r\cos\lambda)\vec{e}_\lambda\}.$

$-\dot\nu^2 r$

$= (\dot\nu\dot\lambda\cos\lambda - \dot\nu\lambda\dot\nu\cos\lambda - \dot\nu^2 r\cos^2\lambda - \dot\nu^2\sin\lambda - \dot\nu\lambda\dot\lambda\sin\lambda - \dot\nu^2 r\sin^2\lambda)\vec{e}_\nu +$

$(\dot\lambda^2 - \dot\lambda^2\nu - \dot\lambda\dot\nu r\cos\lambda + \dot r\dot\nu\sin\lambda + \dot\nu^2\lambda\cos\lambda\sin\lambda - \nu\dot\nu^2\sin^2\lambda)\vec{e}_\nu +$

$(-\dot\lambda\dot\nu - \dot\lambda^2\lambda - \dot\lambda\dot\nu r\sin\lambda - \dot r\dot\nu\cos\lambda - \lambda\dot\nu^2\cos^2\lambda - \nu\dot\nu\cos\lambda\sin\lambda)\vec{e}_\lambda$

$= (\ddot r + 2\dot\nu\dot\lambda\cos\lambda - \dot\nu^2 r - 2\dot\nu^2\sin\lambda + \lambda\ddot\nu\cos\lambda - \lambda\dot\nu\sin\lambda - \nu\ddot\nu\sin\lambda - \nu\dot\nu\cos\lambda$
$- \dot\nu\lambda\nu\cos\lambda - \dot\nu\lambda\dot\lambda\sin\lambda)\vec{e}_\nu +$

$(\ddot\nu + \ddot\lambda\lambda + \dot\lambda^2 + \ddot\nu r\sin\lambda + \dot\nu\dot r\sin\lambda + \dot\nu r\cos\lambda + \dot\lambda^2 - \dot\lambda^2\nu - \dot\lambda\dot\nu r\cos\lambda$
$+ \dot r\dot\nu\sin\lambda + \dot\nu^2\lambda\cos\lambda\sin\lambda - \nu\dot\nu^2\sin^2\lambda)\vec{e}_\nu +$

$(\ddot\lambda - \ddot\lambda\nu + \cancel{\dot\lambda\dot\nu} - \dot r\dot\nu\cos\lambda - r\ddot\nu\cos\lambda + r\dot\nu\sin\lambda - \cancel{\dot\lambda\dot\nu} - \dot\lambda^2\lambda - \dot\lambda\dot\nu r\sin\lambda$
$- \dot r\dot\nu\cos\lambda - \lambda\dot\nu^2\cos^2\lambda - \nu\dot\nu\cos\lambda\sin\lambda)\vec{e}_\lambda$

Newton's Law. $\ddot{\vec r} = -\frac{\mu}{r^3}\vec r + \frac{\vec F}{m}.$

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$e_r : = \ddot r - 2\dot\nu^2\sin\lambda - \dot\nu^2 r + 2\dot\nu\dot\lambda\cos\lambda - \lambda\dot\lambda\dot\nu(\sin\lambda + \cos\lambda)$
$- \nu\ddot\nu\sin\lambda - \nu\dot\nu\cos\lambda + \ddot\nu\lambda\cos\lambda - \dot\nu\lambda\sin\lambda$

$= -\frac{\mu}{r^2} + \frac{F_r}{m}$

$$\boxed{\text{e. } \begin{cases} \ddot r - 2\omega R\dot\nu - 3r\omega^2 = F_r/m. & R=r. \\ R\ddot\nu + 2\omega\dot r = F_\nu/m. & \text{where } \omega = \sqrt{\frac{\mu}{R^3}} \\ R\ddot\lambda + \omega^2 R\lambda = F_\lambda/m. \end{cases}}$$

$e_\nu : \ddot\nu + 2\dot r\dot\nu\sin\lambda + 2\dot\lambda^2 + \ddot\lambda\lambda + r\ddot\nu\sin\lambda - \nu\dot\lambda^2 - r\dot\nu\dot\lambda\cos\lambda + \dot\nu^2\lambda\sin\lambda\cos\lambda$
$- \nu\dot\nu^2\sin^2\lambda + \dot\nu r\cos\lambda = F_\nu/m.$

$e_\lambda : \ddot\lambda - \ddot\lambda\nu + \dot\lambda\dot\nu - r\ddot\nu\cos\lambda - \dot r\dot\nu\cos\lambda + r\dot\nu\sin\lambda - \dot\nu\dot\lambda - \dot\lambda\lambda - r\dot\nu\dot\lambda\sin\lambda$
$- \dot r\dot\nu\cos\lambda - \lambda\dot\nu^2\cos^2\lambda + \nu\dot\nu^2\sin\lambda\cos\lambda = F_\lambda/m.$

$r=R.$ $\nu = \sqrt{\frac{\mu}{R^3}}\,t + c,$ $\lambda = 0 = \dot r = \ddot r = \dot\nu = \ddot\nu = \dot\lambda = \ddot\lambda$ 을 넣어본다

$e_r : -\sqrt{\frac{\mu}{R^3}}R - \sqrt{\frac{\mu}{R^3}}\left(\sqrt{\frac{\mu}{R^3}}t + c\right) = -\frac{\mu}{R^3}$ −5

−10

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$X \equiv \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow \dot{X} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} X + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u.$

1. 0.

$\dot{X} = \underbrace{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}}_{\equiv A} X.$

$\Rightarrow \det|\lambda I - A| = \begin{vmatrix} \lambda & -1 \\ 0 & \lambda \end{vmatrix} = \lambda^2.$

$\Rightarrow$ eigenvalue : $\lambda = 0, 0$

Since both eigenvalues are 0, then the system is not stable.

linear 하게 에러가 쌓이는 시스템.

실제로 $e^{At}$를 구해보면,

$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$

$e^{At} = I + At + \frac{1}{2}(At)^2 + \cdots$

$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & t \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}.$

$X(t) = e^{At}X(0) = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix}$

$\Rightarrow \begin{cases} x_1(t) = x_1(0) + x_2(0)\,t. \\ x_2(t) = x_2(0) \end{cases}$

따라서 stable 하지 않다. ($t$ 가 커지면 $x_1$이 $\infty$로 간다)

system matrix : $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, control matrix $B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$

$\text{rank}\begin{bmatrix} B & AB \end{bmatrix} = \text{rank}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = 2.$

variable 수와 rank[B AB] 가 같으므로 controllable 하다.

$u = \alpha x_1 + \beta x_2 = \begin{bmatrix} \alpha & \beta \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \Lambda X$ 라 하면,

$\dot{X} = AX + Bu = AX + B\Lambda X = (A + B\Lambda)X$

$= \left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ \alpha & \beta \end{bmatrix}\right) X = \begin{bmatrix} 0 & 1 \\ \alpha & \beta \end{bmatrix} X \equiv CX.$

$\det(\lambda I - C) = \begin{vmatrix} \lambda & -1 \\ -\alpha & \lambda - \beta \end{vmatrix} = \lambda(\lambda - \beta) - \alpha = \lambda^2 - \beta\lambda - \alpha.$

$\Rightarrow$ eigenvalues $\lambda = \dfrac{\beta \pm \sqrt{\beta + \alpha}}{2}$

...w consider asymptotically Stability.

definition is, $\lim_{t\to\infty} x(t) = x_e$ where $x_e$ is equilibrium point

$\dot{x} = Cx \Rightarrow X(t) = \exp(Ct)\, X_0$. where $X_0$ is initial state.

$$\begin{bmatrix}0&0\\\alpha&\beta\end{bmatrix}\begin{bmatrix}0&0\\\alpha&\beta\end{bmatrix} = \begin{bmatrix}0&0\\\alpha\beta&\beta^2\end{bmatrix} = C^2$$

$$\begin{bmatrix}0&0\\\alpha&\beta\end{bmatrix}\begin{bmatrix}0&0\\\alpha\beta&\beta^2\end{bmatrix} = \begin{bmatrix}0&0\\\alpha\beta^2&\beta^3\end{bmatrix} = C^3.$$

$$\vdots$$

$$C^n = \begin{bmatrix}0&0\\\alpha\beta^{n-1}&\beta^n\end{bmatrix}.$$

$$\Rightarrow \exp(Ct) = I + \sum_{n=1}^{\infty}\frac{1}{n!}C^n t^n = \begin{bmatrix}1&0\\0&1\end{bmatrix} + \sum_{n=1}^{\infty}\frac{1}{n!}\begin{bmatrix}0&0\\\alpha\beta^{n-1}&\beta^n\end{bmatrix}t^n$$

$$= \begin{bmatrix}1&0\\\frac{\alpha}{\beta}\sum_{n=0}^{\infty}\frac{1}{n!}(\beta t)^n - \frac{\alpha}{\beta} & \sum_{n=0}^{\infty}\frac{1}{n!}(\beta t)^n\end{bmatrix}.$$

$$= \begin{bmatrix}1&0\\\frac{\alpha}{\beta}(e^{\beta t}-1)&e^{\beta t}\end{bmatrix}.$$

$$\Rightarrow X(t) = \begin{bmatrix}1&0\\\frac{\alpha}{\beta}(e^{\beta t}-1)&e^{\beta t}\end{bmatrix}X_0$$

Hence if $\beta \leq 0$, the control-argumented system is asymtotically stable.

$-5$

본인의 controller 만들어야 해서 숫자 명시해야 한다.

...324031



b) the position of the spacecraft $\vec{r} = (\mu - 0.5)\hat{i}$

If the spacecraft is in equilibrium, then EoM for restricted 3 body problem hold with zero velocity and zero acceleration.

$\Rightarrow x = \mu - 0.5$, $y = z = 0 = \dot{x} = \dot{y} = \dot{z} = \ddot{x} = \ddot{y} = \ddot{z}$

① $\ddot{x} - 2\dot{y} - x = -\dfrac{(1-\mu)(x-\mu)}{r_1^3} - \dfrac{\mu(x+1-\mu)}{r_2^3}$

$$\Rightarrow -x = -\frac{(1-\mu)(-0.5)}{0.5^3} - \frac{\mu(0.5)}{0.5^3} = \frac{1}{0.5^2}(1-\mu-\mu)$$

$$\Rightarrow -\mu + 0.5 = \frac{1}{0.5^2}(1-2\mu).$$

Left side : $-\mu + 0.5 = 0.5 - 0.0123 = 0.4877$.

Right side : $\dfrac{1}{0.5^2}(1-2\mu) = \dfrac{1}{0.5^2}(1 - 2\times 0.0123) \approx 3.9016$.

Since 0.4877 and 3.9016 are totally different, the spacecraft is not in equilibrium even the other 2 eqs hold.

(∵) both sides are zero.

Thus, the acceleration $\vec{a}$ is,

$$\ddot{x} = 2\dot{y} + x - \frac{(1-\mu)(x-\mu)}{r_1^3} - \frac{\mu(x+1-\mu)}{r_2^3}$$

$= -0.4877 + 3.9016 = 3.4139$ km/s.

$\Rightarrow \vec{a} = 3.4139\hat{i}$ km/s

To keep the spacecraft in equilibrium, the acceleration should be zero.

$$\Rightarrow \ddot{x} - 2\dot{y} - x = -\frac{(1-\mu)(x-\mu)}{r_1^3} - \frac{\mu(x+1-\mu)}{r_2^3} + \frac{F_x}{m}$$

$$\Rightarrow -x + \frac{(1-\mu)(x-\mu)}{r_1^3} + \frac{\mu(x+1-\mu)}{r_2^3} = \frac{F_x}{m}.$$

$\Rightarrow \dfrac{F_x}{m} = -3.4139$ km/s.

$\therefore \vec{F} = -3.4139\hat{i}$ km/s.

$\ddot{y}$, $\ddot{z}$ 에 관한 식은 이미 추력이 없어도 0이 되므로 안구해도 된다

$\Delta\dot{x} = A\Delta x \Rightarrow \Delta x(t) = e^{At}\Delta x(0) = e^{At}\Delta x_0.$

$= a_1\xi e^{j\omega t} + a_2\bar{\xi}e^{-j\omega t}$

$= a_1\xi(\cos\omega t + j\sin\omega t) + a_2\bar{\xi}(\cos\omega t - j\sin\omega t)$

$= \underline{(a_1\xi + a_2\bar{\xi})\cos\omega t} + \underline{j(a_1\xi - a_2\bar{\xi})\sin\omega t}.$

Since $\Delta x(0) = \Delta x_0$,

$\underline{\Delta x(0) = a_1\xi + a_2\bar{\xi}}$ & $\underline{\Delta\dot{x}(0) = (a_1\xi - a_2\bar{\xi})\omega j}$.

If $a_1 = a_2$, $\Delta x(0) = 2a_1 \mathrm{Re}\,\xi$

If $a_1 = -a_2$, $\Delta x(0) = 2a_1 \mathrm{Im}\,\xi$.

So, the initial condition $\Delta x_0 \in \mathrm{span}\{\mathrm{Re}\,\xi, \mathrm{Im}\,\xi\}$ composes a Halo orbit with a single frequency of $\omega$.